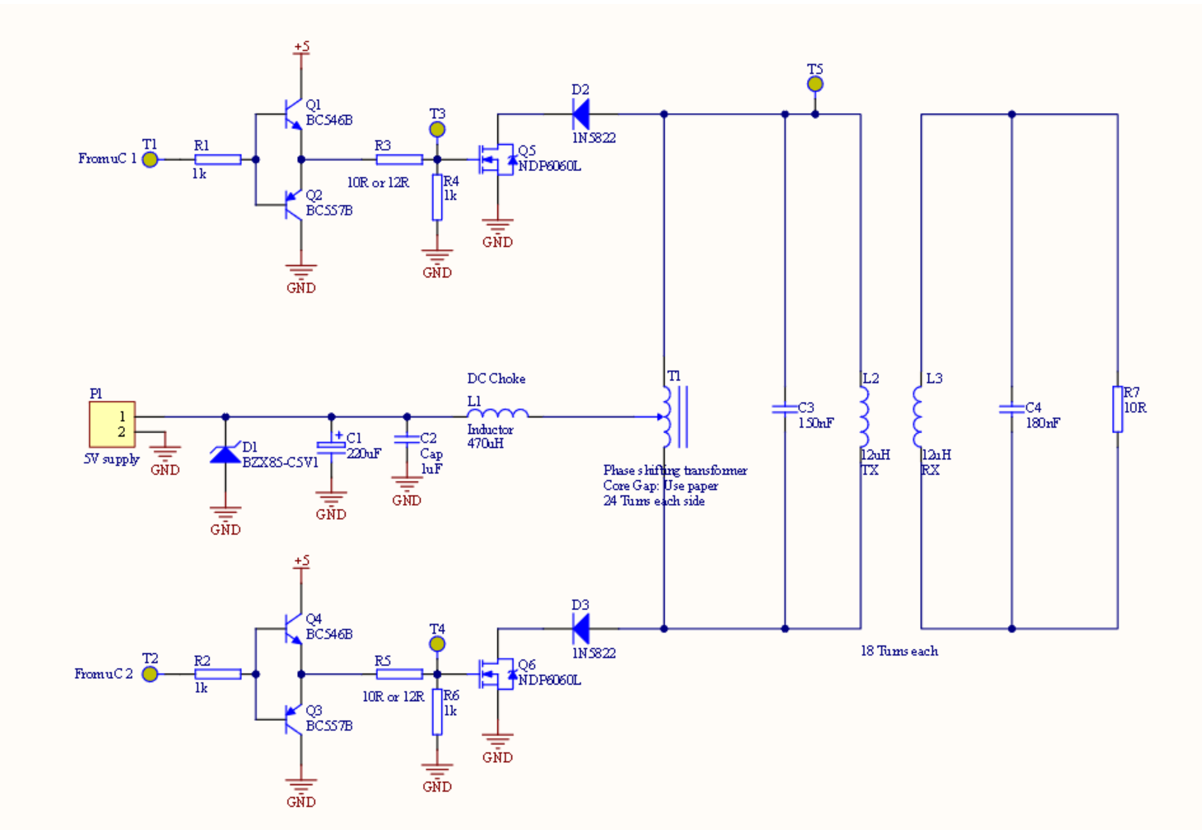
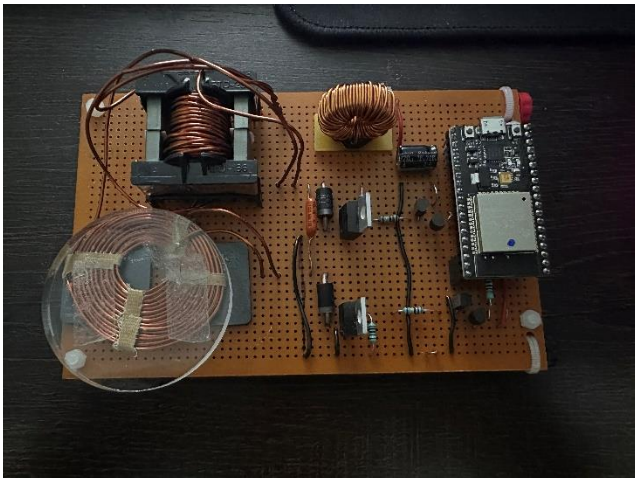
Wireless Power Transfer Circuit Project

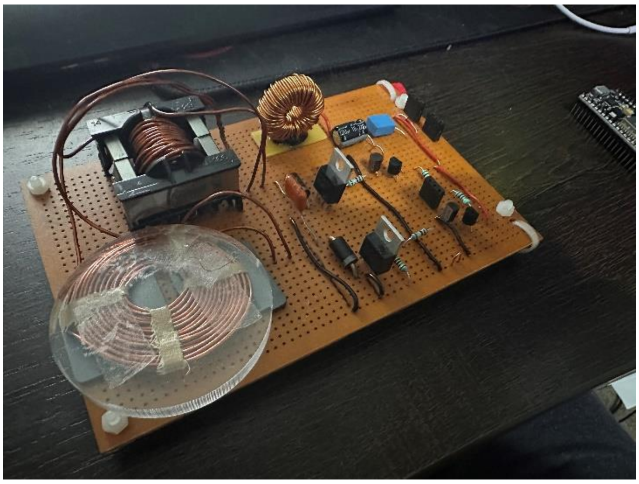
Contents

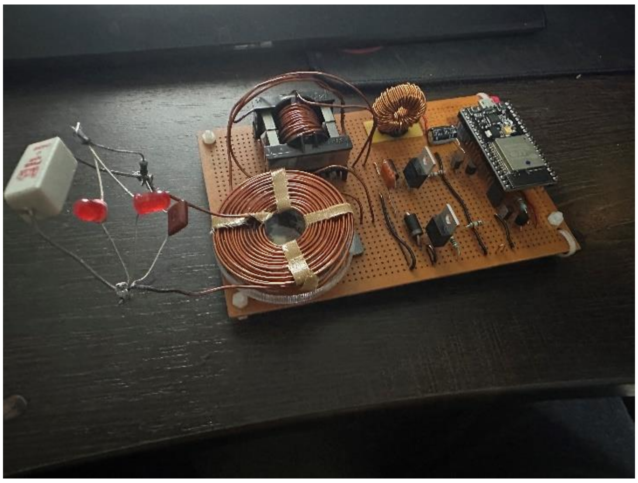
Wireless Power Transfer Schematic

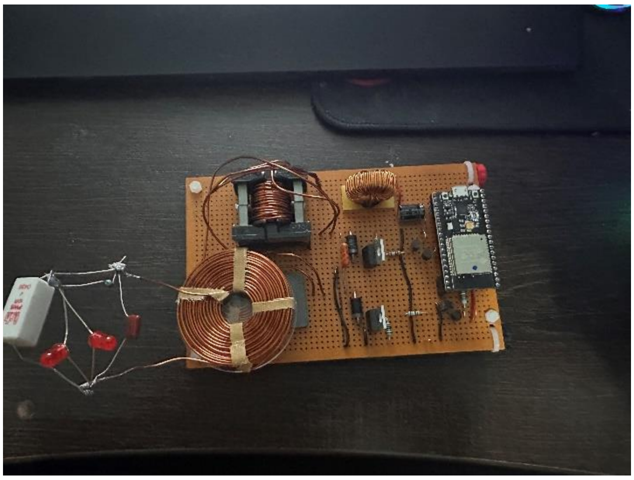


Wireless Power Transfer Full Construction

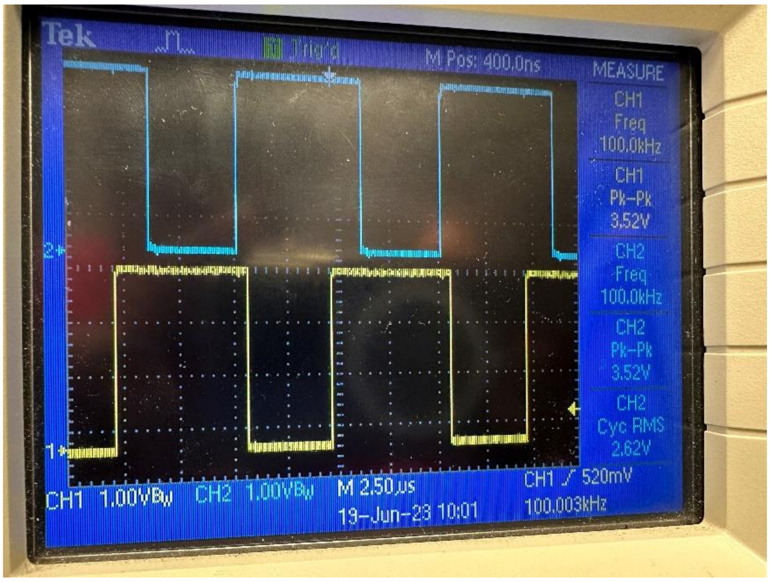








MCU Output Signals (T1 and T2)



Channel 1

Frequency: 100kHZ

Amplitude: 3.56V

Channel 2

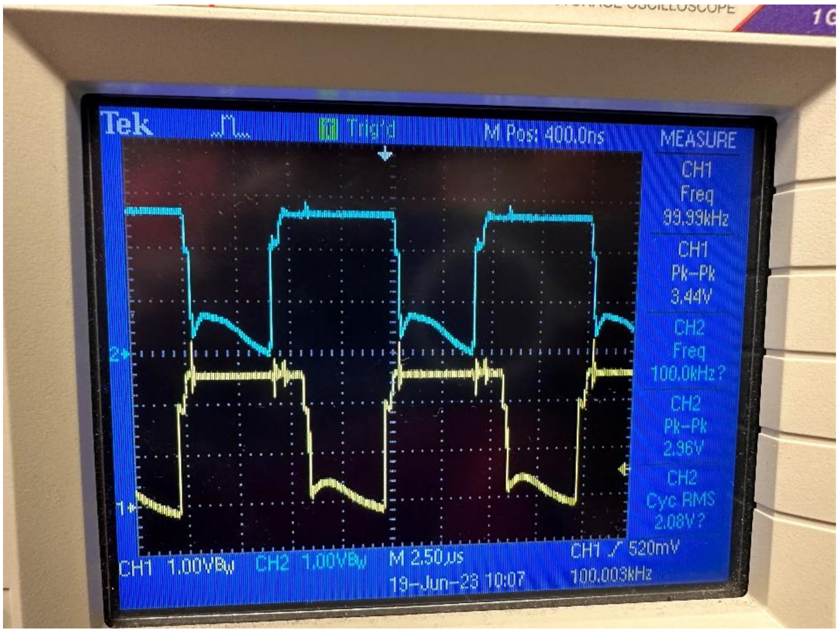
Frequency: 100kHZ

Amplitude: 3.52V

RMS: 2.63V

- Based on the waveforms, I have noticed there is no switching noise.
- Input voltage was 3.3V

MOSFET Gate Signals (T3 and T4)



Channel 1

Frequency: 99.99kHz

Amplitude: 3.44V

Channel 2

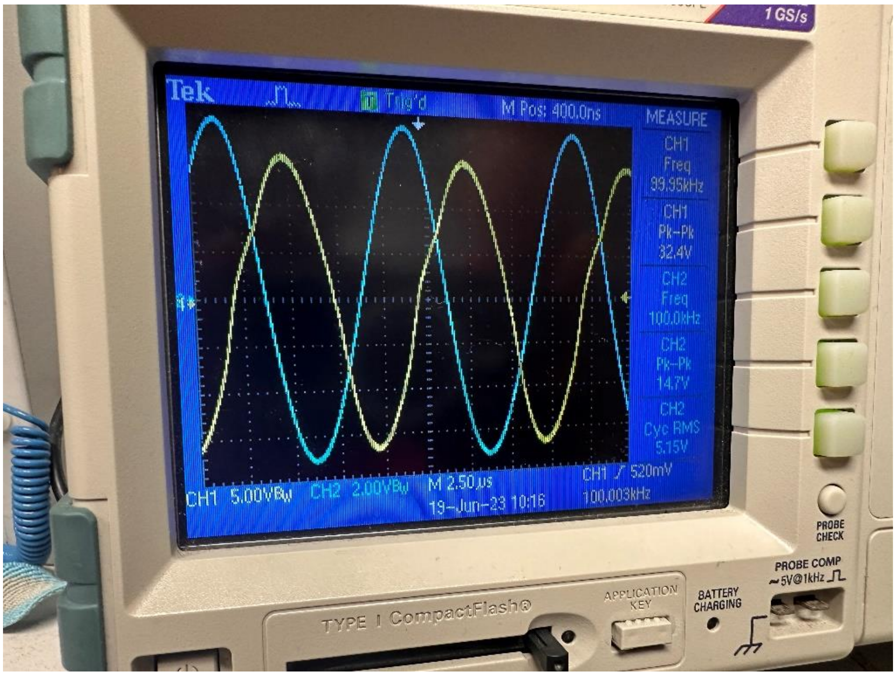
Frequency: 100kHZ

Amplitude: 2.96V

RMS: 2.08V

- Input voltage is 2.6V

Transmission and Receiving Signals (T5)



Channel 1

Frequency: 99.95kHz

Amplitude: 32.4V

Channel 2

Frequency: 100kHZ

Amplitude: 14.7V

RMS: 5.15V

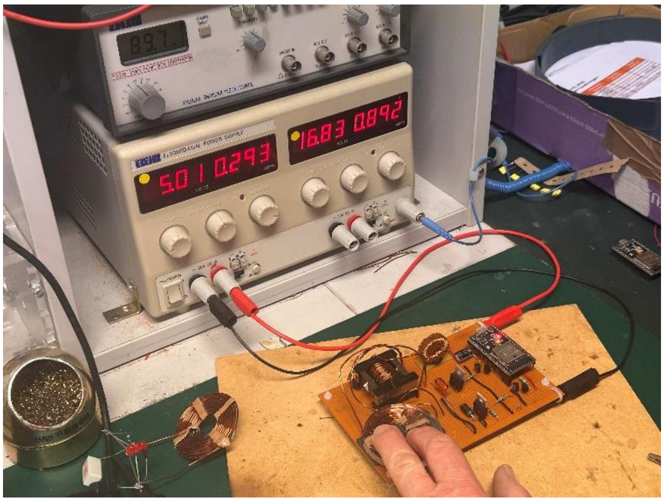
Efficiency

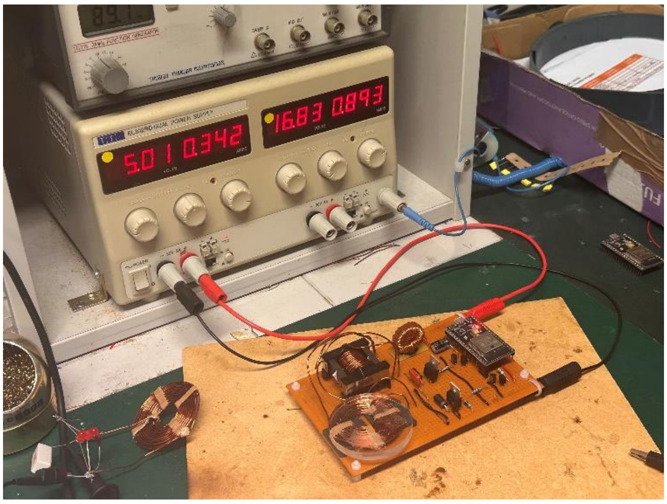
Input Power (power when receiving coils are placed) = 4.88W

Output Power = 2.704 W

Efficiency % = Output Power / Input Power = 2.704 / 4.88 = 56.5%

Offload Current





- Offload Current when pressing on coils is 0.293A
- Offload Current when coils are untouched is 0.342A

Because larger current levels would result in more current being lost during idleness, I was able to reach a low offload current, which is advantageous.